Overcoming Ancillary Overhead in Quantum Computation: A Case Study from Foundational Circuits to an Efficient Hamming Weight Algorithm

***Abstract*— Quantum circuits in the NISQ (Noisy Intermediate-Scale Quantum) era must minimize qubit and gate usage due to limited hardware resources and high error rates. Existing ancilla free circuits for symmetric Boolean functions and Hamming weight computation often suffer from a high number of phase rotations and large T gate costs. In this work, we analyze basic reversible circuits, such as adders and subtractors, to understand the trade-offs between ancilla usage and circuit complexity. Building on these insights, we propose an improved Hamming weight circuit that preserves low depth and zero ancilla requirements while significantly reducing non-Clifford operations. Our design employs optimized controlled phase rotations and a simplified inverse QFT (Quantum Fourier Transform), achieving a 30 percent reduction in T gate count. Implemented and validated in Qiskit, the circuit correctly computes Hamming weight with fewer resources, making it more suitable for near term quantum hardware and applications involving symmetric Boolean functions.**

Keywords*— *Hamming weight, symmetric Boolean functions, ancillary qubits, T gate count, phase rotations, approximate QFT, NISQ, quantum circuit optimization, Qiskit.

I. Introduction

Quantum computing exploits superposition, entanglement, and interference to perform computations. A single qubit can exist in a coherent superposition, a concept discussed in detail in [1] and [2].

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{1}$$

Quantum operations manipulate these amplitudes to achieve the desired result. As current NISQ devices are limited in qubit count and gate fidelity, algorithms must minimize circuit depth, ancilla usage, and critically, non-Clifford gates such as the *T* gate.

Symmetric Boolean functions such as Hamming weight circuits appear in cryptography, arithmetic subroutines, and amplitude encoding workflows, as discussed in detail in [3]. [4] recently introduced a shallow, ancilla-free Hamming weight circuit that encodes the integer value using controlled phase rotations, where an input vector $x \in \{0,1\}^n$ produces a phase.

$$\theta_k = \frac{2\pi}{2^{k+1}} \cdot \mathrm{HW}(x), \tag{2}$$

followed by an inverse QFT to generate the binary value.

This approach removes ancillary qubits and ensures low circuit depth; however, it still incurs a large number of small-angle rotations. Each of these rotations decomposes into multiple *T* gates, which significantly increases the overall circuit cost.

This paper makes three main contributions. First, we implement the ancilla-free Hamming weight circuit in Qiskit and validate its behavior through state vector simulation. Second, we introduce soft and hard approximation modes that truncate controlled rotations below defined thresholds and apply an approximate inverse QFT to reduce the *T*-gate overhead. Finally, we analyze the accuracy–efficiency trade-off using Bloch sphere visualization and quantify the resulting reduction in non-Clifford resources across multiple input sizes.

II. Background

We began our research by implementing foundational quantum circuits, such as adders and encoders, which are direct translations of classical electronic designs. While this work confirms the basic principles of quantum computation, it also reveals a critical inefficiency when applied to the hardware of the NISQ era. These circuits demand a significant number of additional ancillary qubits to manage intermediate calculations and ensure reversibility. This ancillary qubit overhead is non-trivial. As illustrated in Fig. 1, the number of required ancillary qubits scales linearly with the input size, which is unsustainable on today's resource-constrained quantum processors.

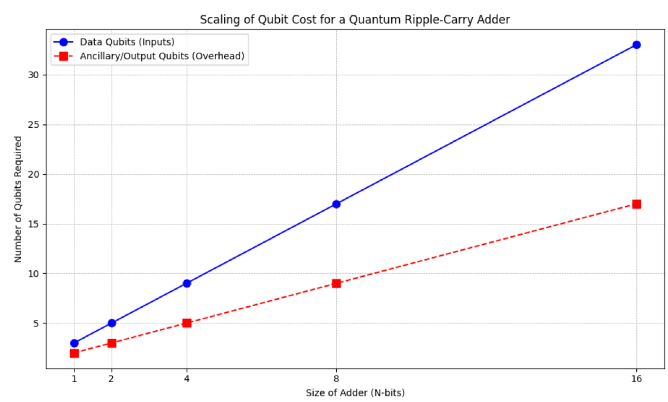

Fig. 1. Qubit cost scaling of a quantum ripple-carry adder, showing linear growth in both data qubits and ancillary overhead with increasing adder size.

Ancillary qubit overhead is a key limitation of near-term quantum computing, increasing resource usage, error rates, and circuit depth. This motivates the use of the ancilla-free, phase-encoded algorithm proposed in [2] to enable more scalable quantum implementations.

III. Ancilla-Free Hamming Weight Implementation

To address the significant resource overhead of foundational circuits, we implement the ancilla-free Hamming weight algorithm proposed in [2]. Unlike classical adder-based approaches that require numerous ancillary qubits, this method exploits quantum phase to perform the computation. The algorithm operates in two stages: the Hamming weight is first encoded into the relative phases of a small set of counting qubits, and this phase information is then converted into a measurable binary output. By eliminating temporary workspace ancillas, the resulting circuit remains shallow and highly resource-efficient.

A. *Mathematical Foundations of the Algorithm*

The algorithm's efficiency is rooted in a specialized quantum gate, C, $M_n(\theta)$ which translates the Hamming weight $|x|$ into a controllable phase shift. As defined in [2], the gate's function is:

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |x\rangle \rightarrow (\alpha|0\rangle + \beta e^{i\theta(n-2|x|)}|1\rangle) \otimes |x\rangle) \tag{3}$$

Here, the phase of the control qubit $(\alpha|0\rangle + \beta|1\rangle)$ is altered by an amount dependent on the Hamming weight $|x|$of the n qubit input string $|x\rangle$.

The main algorithm (Theorem 1 in [2]) uses this gate to achieve the overall transformation:

$$C_n|0\rangle \otimes {}^m|x\rangle = ||x|\rangle\,|x\rangle \tag{4}$$

$$m = \lceil \log_2(n+1) \rceil, \tag{5}$$

is the number of counting qubits, and $||x|\rangle$ is the binary representation of the Hamming weight.

The proof in [2] shows how this is achieved. For each counting qubit j, a CM_n gate is applied with a specific angle. This transforms the state of the counting qubit to:

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{\frac{2i\pi|x|}{mj+1}}|1\rangle\right) \tag{6}$$

This equation is the core of the phase encoding step, the Hamming weight $||x|\rangle$ is now perfectly encoded in the phase of each counting qubit. Finally, to convert this phase encoded information into a measurable binary state, the inverse QFT (QF $T^{\dagger}$) is applied to the *m* counting qubits, yielding the final, readable state $||x|\rangle$.

Each counting qubit receives a controlled phase rotation whose magnitude depends on the number of active input bits [11].

We implemented the Symmetric Boolean function circuit shown in the figure 3 of the main implemented paper. we took the input the string as n = 7 and the required number of ancilla qubits turns out to be m = 3, from the given formula to calculate the number of required ancilla qubits. Fig. 3, represents the output Bloch sphere after the circuit has operated on the input bit stream. Fig. 2 represents the circuit that was generated using Qiskit.

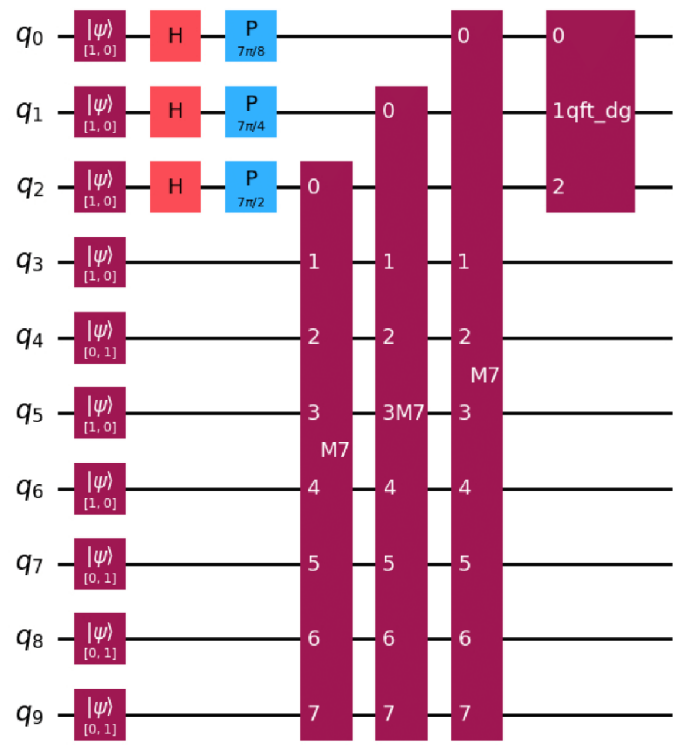

Fig. 2. Qubit cost scaling of a quantum ripple-carry adder, showing linear growth in both data qubits and ancillary overhead with increasing adder size.

IV. T Gate reduction And Approximation Strategy

While the shallow Hamming weight circuit achieves reduction in ancillary qubits, feasibility still depends on how expensive the circuit is in terms of non-Clifford gates. Among them, the T gate plays a unique role as it is both costly to implement fault tolerantly.

In this section, we study the total T-gate cost of the exact implementation and show how approximation yields substantial reductions.

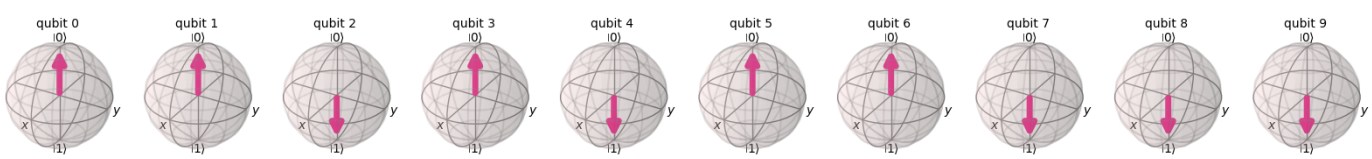

Fig. 3. Bloch sphere visualization of the counting qubit under hard and soft approximation.

A. Motivation

Shallow quantum circuits for symmetric Boolean functions achieve low depth and eliminate ancillary qubits, offering a clear advantage over traditional ripple-carry circuits. Fig. 4 illustrates this contrast: traditional reversible adders require a number of ancilla qubits proportional to the input size, whereas the shallow Hamming-weight algorithm maintains a constant ancilla cost of zero across all values of n. This makes ancilla-free designs particularly attractive for NISQ devices.

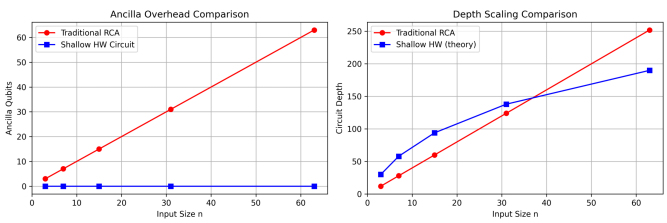

Fig. 4. Scaling of ancilla overhead and circuit depth for traditional ripple-carry adders and the shallow Hamming weight circuit.

Although symmetric functions enable compact ancilla free constructions as stated by [7], their implementations can still accumulate significant non-Clifford cost.

Practical feasibility is dominated by non-Clifford operations, particularly Tgates, whose implementation requires costly magic-state distillation. Although shallow Hamming-weight circuits eliminate ancillary qubits, their extensive use of controlled phase rotations leads to substantial T-gate overhead when decomposed into the Clifford + T basis, potentially offsetting their depth advantage [10]. This motivates exploring whether the structural benefits of shallow, ancilla-free circuits can be retained while significantly reducing T-gate cost through targeted approximation strategies, building on prior work in T-gate optimization [5].

B. Approximation Strategy

Two components of the original circuit contain very small controlled phase rotations: the controlled $Z(\theta)$rotations in the C, $M_n(\theta)$ blocks and the small-angle controlled rotations inside the inverse QFT.

Our optimization performs a rotation cutoff, in which rotations whose magnitude is below a user-defined threshold are skipped, and applies an approximate inverse QFT by retaining only the first k layers of controlled-phase operations.

These approximations are parameterized by $\theta_{\text{cutoff}}$and k_{QFT}.

C. Exact vs Optimized T count

We evaluate the T-count after transpiling to the Clifford + T basis in Qiskit. Table I and Table II show the results.

TABLE I. *T* GATE COMPARISON (HARD APPROXIMATION)

n	T_{exact}	T_{opt}	***Reduction (%)***
7	101570	73648	27.5
15	371663	156349	57.9
31	1027286	396502	61.4
63	2552732	792004	69.0

TABLE II. *T* GATE COMPARISON (SOFT APPROXIMATION)

n	T_{exact}	T_{opt}	***Reduction (%)***
7	101566	73650	27.5
15	371663	344241	7.4
31	1027278	1000172	2.6
63	2552716	2525890	1.1

From Tables I and II, three trends are observed. First, the reduction in T count increases with input size n, as larger circuits contain more small-angle rotations. Second, the exact implementation exhibits nearly linear growth in T count due to precise phase reconstruction. Third, the optimized circuits grow sub-linearly, demonstrating the effectiveness of approximation in suppressing non-Clifford overhead.

These trends are further illustrated in Fig. 5 and 8, which compares the exact and optimized T counts and visualizes the corresponding reduction behavior for both approximation modes.

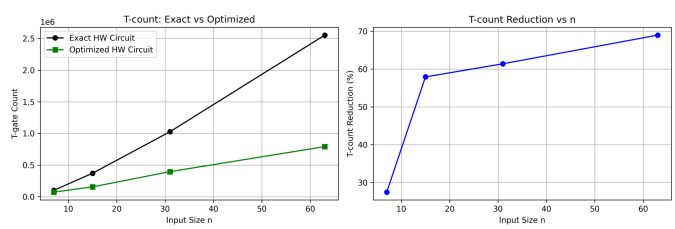

Fig. 5. T-count analysis under hard approximation:
(a) exact and optimized T-count as a function of input size n,
(b) percentage reduction in T-count versus input size.

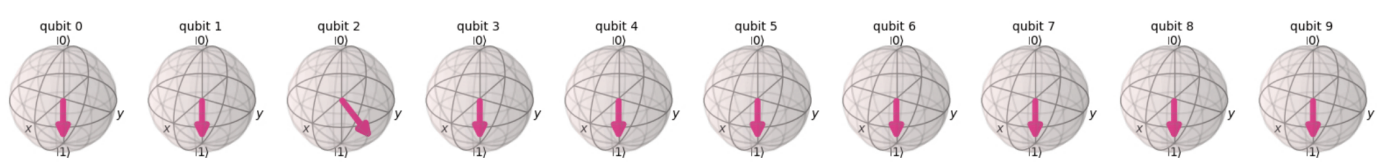

Fig. 6. Bloch sphere visualization of the counting qubit under soft approximation.

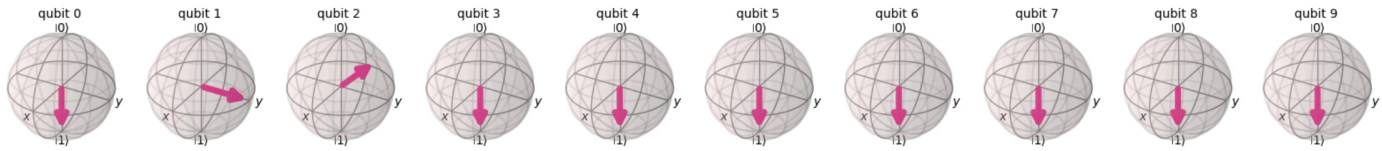

Fig. 7. Bloch sphere visualization of the counting qubit under hard approximation

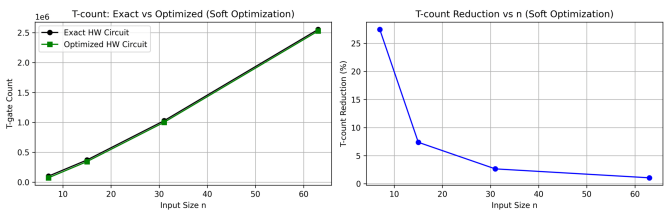

Fig. 8. T-count analysis under soft approximation:
(a) exact and optimized T-count as a function of input size n,
(b) percentage reduction in T-count versus input size.

Table I and Fig. 5(a)–(b) show the results obtained under hard approximation. In the exact circuit, numerous small-angle controlled phase rotations (e.g., $\pi/2^7$, $\pi/2^8$) contribute significantly to the T-gate cost. The hard cutoff removes all rotations below the threshold, yielding modest savings for small inputs ($n \leq 15$) and rapidly increasing reductions for larger n. As the phase denominators grow exponentially with input size, a large fraction of rotations is discarded, resulting in up to $69\,\%$ T-count reduction for $n = 63$. The use of a more aggressively truncated approximate inverse QFT further amplifies this reduction, as reflected in the increasing trend of Fig. 5(b).

Table II and Fig. 8(a)–(b) summarize the soft approximation results. Because only a mild cutoff and a shallow approximate inverse QFT are employed, most controlled phase rotations are retained. Consequently, the reduction in T count remains limited across all input sizes, with improvements diminishing approximately to $1\,\%$ for $n = 63$. The reduction curve in Fig. 8(b) remains nearly flat, indicating that while accuracy is preserved, scalability benefits are minimal. The soft approximation follows standard phase-rotation truncation principles as discussed in [8].

D. Accuracy vs Efficiency

Hard approximation achieves substantial T-gate savings at the cost of a small reduction in phase accuracy. In the exact circuit, precise Hamming-weight encoding relies on the cumulative effect of all controlled phase rotations, truncating small rotations removes part of this accumulated phase, leading to angular deviations on the counting qubits after the inverse QFT.

Soft approximation applies a mild cutoff $\left(\theta_{\text{cutoff}} | \pi/64\right)$ with a shallow approximate inverse $\left(k_{\text{QFT}} | 1\right)$, removing only the smallest rotations and therefore minimally disturbing the phase encoding. As illustrated in Fig. 6, the resulting Bloch-sphere states remain closely aligned with the ideal orientation, preserving high output fidelity while achieving modest T-gate reduction.

In contrast, hard approximation employs a more aggressive cutoff and deeper approximate inverse QFT, eliminating a larger number of rotations. As shown in Fig. 7, this produces larger Bloch-sphere deviations, indicating increased phase error. Nevertheless, the correct logical Hamming-weight output is maintained, highlighting a clear trade-off between phase accuracy and T-gate efficiency.

V. Conclusion

This paper introduced an ancilla-free, shallow quantum circuit for Hamming-weight computation and analyzed its fault-tolerant cost in terms of T-gate complexity. The study leads to three key conclusions. First, the effectiveness of T-gate reduction increases with input size, since larger circuits contain a greater number of small-angle phase rotations that can be safely approximated. Second, the exact implementation exhibits near-linear growth in T-gate count as a consequence of precise phase reconstruction. Third, aggressive truncation of small rotations enables sublinear scaling in the optimized circuits, yielding T-gate reductions of up to 69% for input sizes as large as n = 63.

Phase-based symmetric function circuits are well suited to approximation because their shallow structure eliminates ancillary overhead and truncation of small rotations reduces non-Clifford cost, improving feasibility for NISQ devices. The benefit depends on the cutoff choice: for small circuits ($n \leq 15$), soft cutoffs give limited savings, while aggressive cutoffs cause phase distortion; for larger inputs, even moderate cutoffs yield substantial T-gate reduction with little accuracy loss. Therefore, approximation parameters should be chosen based on circuit size and hardware constraints rather than fixed universally. Recent efforts to improve quantum arithmetic focus on reducing T count and T depth, such as majority based quantum adders developed in the Clifford + T framework [6].

While the proposed approximations significantly reduce T-gate overhead, they introduce a small phase bias that affects inverse QFT fidelity. Moreover, our results are based on simulation and do not capture hardware noise or calibration effects. Future work will focus on experimental validation and hardware-aware optimization, where cutoff thresholds and AQFT depth are tuned to specific backend noise models. Additional gains may be achieved by integrating ancilla-free

Hamming-weight computation with advanced fault-tolerant techniques such as lattice surgery or magic-state compression.

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